Neurobiological Basis of Intelligence

Homework 1 — Computing Architectures: From Neurons to Nodes

Instructions: Questions marked with an asterisk ($*$) are advanced. **Due date:** March 23.

Problem 1: Propagation of Spiking Activity in Feedforward Networks

This problem explores how activity propagates across a chain of neuronal populations and connects that biological mechanism to information flow in deep artificial networks.

Model Setup

Consider a feedforward network of leaky integrate-and-fire neurons with K layers and N neurons per layer. Every neuron in layer k receives equal excitatory input from all neurons in layer $k-1$, with synaptic coupling J.

Neurons in layer 1 receive suprathreshold external drive and fire asynchronously at rate ν_1. Neurons in layers $k > 1$ receive a shared subthreshold constant current I_0 in addition to feedforward input.

For $k > 1$, model each neuron as:

$$\tau_m \frac{dV_i^{(k)}(t)}{dt} = -V_i^{(k)}(t) + I_0 + I_{i,\mathrm{syn}}^{(k)}(t).$$

When $V_i^{(k)}$ reaches threshold 1, a spike is emitted and the voltage is reset to 0:

$$V_i^{(k)}(t^-) = 1 \;\Rightarrow\; V_i^{(k)}(t^+) = 0.$$

A single presynaptic spike at time t_s contributes synaptic current:

$$\Delta I_{\mathrm{syn}}(t) = J\, e^{-(t-t_s)/\tau_s} H(t - t_s),$$

where τ_s is the synaptic decay time constant and $H(\cdot)$ is the Heaviside step function.

Use the following assumptions throughout:

- $N = 1000$.
- ν_1 ranges from 0 to 1000 Hz.
- Neurons in all layers fire asynchronously (briefly justify this approximation).

1) Activity in the Second Layer

For each value of J (in seconds):

$$J \in \{0.08,\, 0.16,\, 0.202\}.$$

a) Compute the second-layer firing rate ν_2 as a function of ν_1, and plot $\nu_2(\nu_1)$.

b) Determine the critical value of ν_1 for which ν_2 first becomes nonzero.

c) For which values of J above are there values of ν_1 such that $\nu_2 > \nu_1$?

d) Find (numerically, or preferably analytically) the critical value of J above which there exist values of ν_1 for which $\nu_2 > \nu_1$.

2) Activity in Deep Layers

Compute ν_k for deep layers ($k \gg 1$) using one or both approaches below.

a) **Numerical approach:** Compute ν_k iteratively across layers for different ν_1 values (e.g., up to $k = 30$). Compare plots of ν_k vs. ν_1 for $k = 2, 10, 20, 30$.

b) **(*) Analytical approach:** Construct the map $\nu_k = F(\nu_{k-1})$ and analyze it as a discrete-time iterated map where layer index k is the time step. Find fixed points and determine their stability (see Strogatz, *Nonlinear Dynamics and Chaos*, Ch. 10).

c) Determine exact boundaries between qualitatively different J regimes: low, intermediate, and high J.

d) Determine how deep-layer activity depends on ν_1. Interpreting ν_1 as sensory stimulus intensity (e.g., visual contrast), discuss implications for transmitting stimulus intensity through feedforward chains.

e) **(*) Food for thought:** Propose an alternative representation of input-layer stimulus intensity (for example, one relying on synchronous activation) that could propagate with higher fidelity than asynchronous rate coding.

Problem 2: Dendritic Computation

Required reading for Problem 2: Read *Poirazi & Mel (2003)* and *Poirazi & Mel (2025)*. Both papers are provided in the `Papers` folder of the course repository.

Warm Up: The Perceptron vs. The Two-Stage Nonlinear Model

To demonstrate how internal neuronal architecture (point vs. branched) dictates computational capacity.

a) **The Linear Bottleneck:** Consider a standard McCulloch-Pitts point neuron $y = H(w_1 x_1 + w_2 x_2 + b)$, where H is the Heaviside step function. Prove algebraically that no set of weights (w_1, w_2) and bias b can solve the **XOR problem**. Provide a 2D sketch illustrating the lack of a linear separator for the XOR input coordinates.

b) **The Two-Stage Solution:** Now consider a neuron with two dendritic branches (B_1, B_2) feeding into a soma.

- B_1 (OR-like): $v_1 = 1$ if $(x_1 + x_2 \geq 1)$, else 0.
- B_2 (AND-like): $v_2 = 1$ if $(x_1 + x_2 \geq 2)$, else 0.
- Soma: $y = H(w_{B1}v_1 + w_{B2}v_2 + b_{soma})$.

Find a set of weights (w_{B1}, w_{B2}) and b_{soma} that allows this single neuron to solve XOR. Compare this to a standard Multi-Layer Perceptron (MLP); which biological component is effectively acting as the "hidden layer"?

Scaling to d-ANNs (Chavlis & Poirazi, 2025)

Analyze the efficiency and robustness of Dendritic Artificial Neural Networks (d-ANNs).

a) **Structural Connectivity & Efficiency:** Compare the Restricted Sampling of d-ANNs with the architecture of Convolutional Neural Networks (CNNs) and randomly sparse ANNs. Calculate the parameter savings if a 1,000-input layer transitions from "Fully Connected" to a d-ANN model where each neuron has 5 branches, each sampling only 5% of the input space.

b) **The Diversity Strategy:** Chavlis & Poirazi observe that d-ANN nodes often respond to **multiple classes**, unlike classical ANNs that strive for class-specificity. Based on the paper, explain why this "diversity strategy" makes the network more robust to noise and overfitting.

c) **Critical Evaluation:** Discuss the trade-offs of d-ANNs. While they are parameter-efficient, what are the potential disadvantages regarding computational complexity (simulation time) and biological plausibility (specifically regarding the use of Backpropagation to train dendritic structures)?